INTERNATIONAL SEARCH REPORT

International application No.
PCT/SG2021/050690

A. CLASSIFICATION OF SUBJECT MATTER

G01N 33/68 (2006.01) G01N 33/564 (2006.01)

According to International Patent Classification (IPC)

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01N C12Q

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

FAMPAT/EMBASE/BIOSIS/MEDLINE: SSB, TROVE2, ZHX2, adalimumab, rheumatoid arthritis, autoantibody, and related terms

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X Y	Immunome™. 4 July 2018 [Retrieved on 2022-01-19 from https://hoc.bme.uh.edu/services/autoantibody-screening-autoantigen-arrays/immunome/] Link to Excel spreadsheet containing all >1600 targets of the IMMUNOME™ v4 Discovery Array	15-18 13-14
X A	ORTEA I. ET AL., Independent Candidate Serum Protein Biomarkers of Response to Adalimumab and to Infliximab in Rheumatoid Arthritis: An Exploratory Study. *PLoS One.*, 6 April 2016, Vol. 11, No. 4, pages e0153140 (1-13) [Retrieved on 2022-01-19] <DOI: 10.1371/JOURNAL.PONE.0153140> Whole document	19 1-12

☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

*Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"D" document cited by the applicant in the international application

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
19/01/2022 *(day/month/year)*	*31/01/2022* *(day/month/year)*

Name and mailing address of the ISA/SG	Authorized officer
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INTERNATIONAL SEARCH REPORT

International application No.
PCT/SG2021/050690

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X A	MATSUDAIRA R. ET AL., Anti-Ro/SSA antibodies are an independent factor associated with an insufficient response to tumor necrosis factor inhibitors in patients with rheumatoid arthritis. *J Rheumatol.*, 1 October 2011, Vol. 38, No. 11, pages 2346-2354 [Retrieved on 2022-01-19] <DOI: 10.3899/JRHEUM.101295.> Whole document	19 1-12
Y	WO 03/064656 A1 (SENSE PROTEOMIC LIMITED) 7 August 2003 Page 3 lines 5-11 and Examples 1-2	13-14
P,X	CHEN P. K. ET AL., Anti-TROVE2 Antibody Determined by Immune-Related Array May Serve as a Predictive Marker for Adalimumab Immunogenicity and Effectiveness in RA. *J Immunol Res.*, 8 March 2021, Vol. 2021, pages 6656121 (1-13) [Retrieved on 2022-01-19] <DOI: 10.1155/2021/6656121.> Whole document	1-12 and 15-19
P,A	MORI A. ET AL., Presence of anti-nuclear antibodies is a risk factor for the appearance of anti-drug antibodies during infliximab or adalimumab therapy in patients with rheumatoid arthritis. *PLoS One.*, 14 December 2020, Vol. 5, No. 12, pages e0243729 (1-16) [Retrieved on 2022-01-19] <DOI: 10.1371/JOURNAL.PONE.0243729> Whole document	-

INTERNATIONAL SEARCH REPORT

International application No.
PCT/SG2021/050690

Box No. I Nucleotide and/or amino acid sequence(s) (Continuation of item 1.c of the first sheet)

1. With regard to any nucleotide and/or amino acid sequence disclosed in the international application, the international search was carried out on the basis of a sequence listing:

   a. ☐ forming part of the international application as filed:

      ☐ in the form of an Annex C/ST.25 text file.

      ☐ on paper or in the form of an image file.

   b. ☐ furnished together with the international application under PCT Rule 13*ter*.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.

   c. ☐ furnished subsequent to the international filing date for the purposes of international search only:

      ☐ in the form of an Annex C/ST.25 text file (Rule 13*ter*.1(a)).

      ☐ on paper or in the form of an image file (Rule 13*ter*.1(b) and Administrative Instructions, Section 713).

2. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.

3. Additional comments:

   Although a sequence listing has been filed or furnished, it was not used for the purposes of this search.

   Since only one version or copy of a sequence listing has been filed or furnished, the statements under item 2 are not required.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/SG2021/050690

Note: This Annex lists known patent family members relating to the patent documents cited in this International Search Report. This Authority is in no way liable for these particulars which are merely given for the purpose of information.

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 03/064656 A1	07/08/2003	AU 2003238441 B2	30/10/2008
		WO 03064656 A1	07/08/2003
		ES 2268376 T3	16/03/2007
		EP 1470229 A1	27/10/2004
		DE 60305643 T2	03/05/2007
		CA 2474457 A1	07/08/2003
		JP 2005516074 A	02/06/2005
		DK 1470229 T3	02/10/2006
		AT 328092 T	15/06/2006
		US 2005/0221308 A1	06/10/2005